Optimization — Homework 3

Solutions

Mitchel Carson

September 19, 2026

Conventions. Vectors are columns; inequalities between vectors are componentwise; A_j is the j-th column of A and a_i^{T} its i-th row. For a minimisation LP I use the standard primal–dual dictionary

Primal (min $c^{\mathsf{T}}x$)	**Dual** (max $p^{\mathsf{T}}b$)
constraint $a_i^{\mathsf{T}}x \geq b_i$	$p_i \geq 0$
constraint $a_i^{\mathsf{T}}x \leq b_i$	$p_i \leq 0$
constraint $a_i^{\mathsf{T}}x = b_i$	p_i free
variable $x_j \geq 0$	$p^{\mathsf{T}}A_j \leq c_j$
variable $x_j \leq 0$	$p^{\mathsf{T}}A_j \geq c_j$
variable x_j free	$p^{\mathsf{T}}A_j = c_j$

and the three facts *weak duality* ($p^{\mathsf{T}}b \leq c^{\mathsf{T}}x$ for any feasible pair), *strong duality* (if either problem has an optimal solution so does the other, with equal values), and *complementary slackness* (a feasible pair is optimal iff $p_i(a_i^{\mathsf{T}}x - b_i) = 0$ for all i and $(c_j - p^{\mathsf{T}}A_j)x_j = 0$ for all j).

Problem 1. Computing a dual

The LP is

$$\begin{array}{lrcrcrcrcl} \min & x_1 & - & x_2 & & & & & & \\ \text{s.t.} & 2x_1 & + & 3x_2 & - & x_3 & + & x_4 & \leq & 0 \\ & 3x_1 & + & x_2 & + & 4x_3 & - & 2x_4 & \geq & 3 \\ & -x_1 & - & x_2 & + & 2x_3 & + & x_4 & = & 6 \\ & x_1 \leq 0, & & x_2, x_3 \geq 0, & & & & x_4 \text{ free.} & & \end{array}$$

In matrix form: $c = (1, -1, 0, 0)^{\mathsf{T}}$, $b = (0, 3, 6)^{\mathsf{T}}$, $A = \begin{pmatrix} 2 & 3 & -1 & 1 \\ 3 & 1 & 4 & -2 \\ -1 & -1 & 2 & 1 \end{pmatrix}$. Note that x_4 carries no sign restriction, so it is a free variable.

Derivation via the Lagrangian. Attach a multiplier p_i to each of the three general constraints and keep the sign constraints on x explicit. Define

$$g(p) = \min_{x_1 \leq 0,\, x_2, x_3 \geq 0,\, x_4 \in \mathbb{R}} \left[c^{\mathsf{T}}x + p^{\mathsf{T}}(b - Ax) \right] = p^{\mathsf{T}}b + \min_{x_1 \leq 0,\, x_2, x_3 \geq 0,\, x_4 \in \mathbb{R}} \sum_{j=1}^{4} (c_j - p^{\mathsf{T}}A_j)x_j.$$

Signs of p. We want $g(p)$ to be a lower bound on the primal optimum, i.e. $p^\mathsf{T}(b - Ax) \le 0$ for every primal-feasible x. Row 1 has $b_1 - a_1^\mathsf{T} x \ge 0$, so we need $p_1 \le 0$; row 2 has $b_2 - a_2^\mathsf{T} x \le 0$, so $p_2 \ge 0$; row 3 has $b_3 - a_3^\mathsf{T} x = 0$, so p_3 is free.

Constraints on p. The inner minimum is $-\infty$ unless each term is bounded below over the allowed sign of x_j: $x_1 \le 0$ requires $c_1 - p^\mathsf{T} A_1 \le 0$; $x_2, x_3 \ge 0$ require $c_j - p^\mathsf{T} A_j \ge 0$; x_4 free requires $c_4 - p^\mathsf{T} A_4 = 0$. When these hold the minimum is 0 and $g(p) = p^\mathsf{T} b = 0p_1 + 3p_2 + 6p_3$. Maximising the lower bound gives the dual, in agreement with the dictionary above.

Answer.

$$
\begin{array}{lrcrcrcll}
\max & 3p_2 + 6p_3 & & & & & & & \\
\text{s.t.} & 2p_1 & + & 3p_2 & - & p_3 & \ge & 1 & (x_1 \le 0) \\
& 3p_1 & + & p_2 & - & p_3 & \le & -1 & (x_2 \ge 0) \\
& -p_1 & + & 4p_2 & + & 2p_3 & \le & 0 & (x_3 \ge 0) \\
& p_1 & - & 2p_2 & + & p_3 & = & 0 & (x_4 \text{ free}) \\
& p_1 \le 0, & & p_2 \ge 0, & & p_3 \text{ free.} & & & \\
\end{array}
$$

(Equivalently, substituting $q_1 = -p_1 \ge 0$ gives a version with all sign-constrained multipliers nonnegative.)

Going further: what are the optimal values? The problem only asks for the dual, but it is instructive to solve the pair.

The dual is infeasible. The equality gives $p_3 = 2p_2 - p_1$. Substituting into the third constraint: $-p_1 + 4p_2 + 2(2p_2 - p_1) = -3p_1 + 8p_2 \le 0$. But $p_1 \le 0$ and $p_2 \ge 0$ make both $-3p_1$ and $8p_2$ nonnegative, so $p_1 = p_2 = 0$ and hence $p_3 = 0$. Then the first constraint reads $0 \ge 1$, which is false.

The primal is feasible and unbounded. The point $x^0 = (0, 0, 2, 2)$ is feasible: the rows evaluate to $0 \le 0$, $4 \ge 3$, $6 = 6$. The direction $d = (-1, 0, 1, -3)$ satisfies $Ad = (-6,\ 7,\ 0)$, $d_1 \le 0$, $d_2, d_3 \ge 0$, so $x^0 + \theta d$ is feasible for every $\theta \ge 0$, while the cost is $c^\mathsf{T}(x^0 + \theta d) = -\theta \to -\infty$ (Figure 1).

This is exactly what duality predicts: by weak duality any dual-feasible p would give a finite lower bound on the primal, so an unbounded primal *forces* an infeasible dual. Both facts are confirmed numerically in Appendix A.

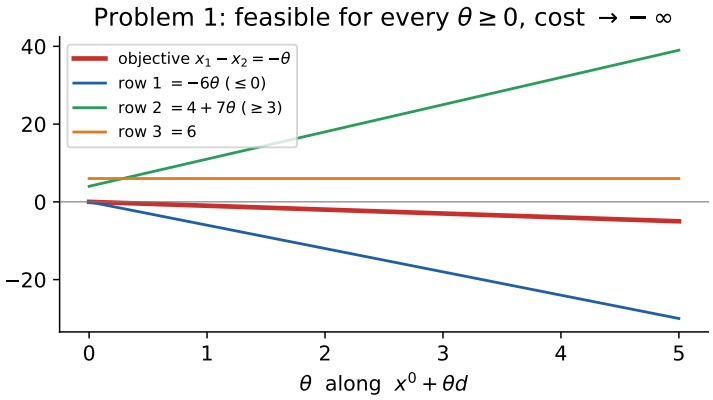


Figure 1: Along $x^0 + \theta d$ all three constraints remain satisfied while the objective decreases without bound.

Problem 2. A theorem of the alternative

For a matrix $A \in \mathbb{R}^{m \times n}$ we show the equivalence of

(i) $Ax \geq 0$ and $x \geq 0$ imply $x_1 = 0$;

(ii) there exists $p \geq 0$ with $p^{\mathsf{T}} A \leq 0$ and $p^{\mathsf{T}} A_1 < 0$.

Proof of (ii) $\Rightarrow$ (i). Let p be as in (ii) and let x satisfy $Ax \geq 0$, $x \geq 0$. Since $p \geq 0$ and $Ax \geq 0$, we have $p^{\mathsf{T}} Ax \geq 0$. On the other hand

$$p^{\mathsf{T}} Ax = \sum_{j=1}^{n} (p^{\mathsf{T}} A_j)\, x_j,$$

and every term is ≤ 0 because $p^{\mathsf{T}} A_j \leq 0$ and $x_j \geq 0$. A sum of nonpositive terms that is ≥ 0 must have every term equal to zero; in particular $(p^{\mathsf{T}} A_1) x_1 = 0$. Since $p^{\mathsf{T}} A_1 < 0$ strictly, $x_1 = 0$. □

Proof of (i) $\Rightarrow$ (ii). Consider the linear program

$$\text{(P)} \qquad \min\ -x_1 \quad \text{s.t.} \quad Ax \geq 0, \quad x \geq 0.$$

It is feasible ($x = 0$). Statement (i) says every feasible x has objective value $-x_1 = 0$, so (P) has an optimal solution, with optimal value 0. Using the dictionary (constraints "$\geq$" give $p \geq 0$; variables $x \geq 0$ give $p^{\mathsf{T}} A_j \leq c_j$; here $c = -e_1$ and the right-hand side is $b = 0$), the dual is

$$\text{(D)} \qquad \max\ p^{\mathsf{T}} 0 = 0 \quad \text{s.t.} \quad p \geq 0, \quad p^{\mathsf{T}} A_1 \leq -1, \quad p^{\mathsf{T}} A_j \leq 0 \ \ (j = 2, \ldots, n).$$

By strong duality, since (P) has an optimal solution, so does (D); in particular (D) is *feasible*. Any feasible p satisfies $p \geq 0$, $p^{\mathsf{T}} A \leq 0$ and $p^{\mathsf{T}} A_1 \leq -1 < 0$, which is (ii). □

Answer. (i) $\Leftrightarrow$ (ii). The easy direction is a one-line "multiply and add" argument; the hard direction is strong LP duality applied to $\min\{-x_1 : Ax \geq 0,\ x \geq 0\}$, whose dual is a pure feasibility problem.

Remark on the strict inequality. The dual produces $p^{\mathsf{T}} A_1 \leq -1$, which looks stronger than $p^{\mathsf{T}} A_1 < 0$, but the two are equivalent: the conditions $p \geq 0$, $p^{\mathsf{T}} A \leq 0$ are homogeneous, so any p with $p^{\mathsf{T}} A_1 < 0$ can be rescaled to $p / |p^{\mathsf{T}} A_1|$, which has $p^{\mathsf{T}} A_1 = -1$. This rescaling trick is the standard way to convert a strict inequality (which LPs cannot express) into a non-strict one.

Remark on the structure. If (i) fails there is a feasible x with $x_1 > 0$; then θx is feasible for all $\theta \geq 0$, (P) is unbounded, and weak duality forces (D) to be infeasible, so (ii) fails. Thus exactly one of "there is $x \geq 0$ with $Ax \geq 0$, $x_1 > 0$" and "(ii)" holds, which is the usual form of a theorem of the alternative.

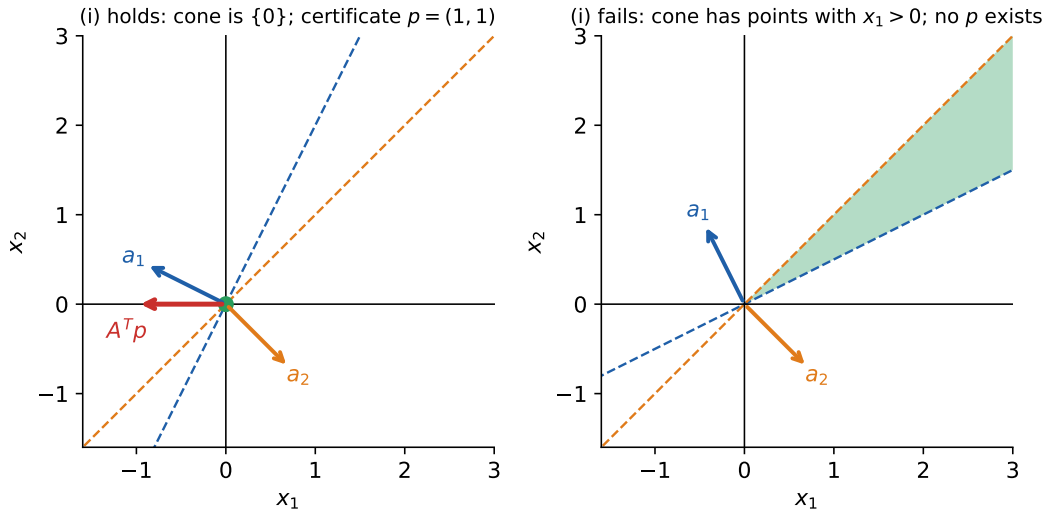


Figure 2: Green: the cone $\{x \geq 0,\ Ax \geq 0\}$; dashed lines: $a_i^{\mathsf{T}} x = 0$. Left, $A = \left(\begin{smallmatrix} -2 & 1 \\ 1 & -1 \end{smallmatrix}\right)$: the cone is $\{0\}$ and $p = (1, 1)$ gives $A^{\mathsf{T}} p = (-1, 0)$, nonpositive with a strictly negative first entry. Right, $A = \left(\begin{smallmatrix} -1 & 2 \\ 1 & -1 \end{smallmatrix}\right)$: the cone contains points with $x_1 > 0$, e.g. $x = (1.5, 1)$, so no certificate p can exist.

Numerical check. For 2000 random integer matrices I tested (i) and (ii) independently by LP; they agreed in every case (1140 both true, 860 both false, 0 disagreements; Appendix A).

Problem 3. An infeasible system has a small infeasible subsystem

Let $A \in \mathbb{R}^{m \times n}$ with $m = 500$, $n = 100$, and suppose $\{x : Ax \leq b\} = \emptyset$. We show that some subset of at most $n + 1 = 101$ of the inequalities $a_i^{\mathsf{T}} x \leq b_i$ is already infeasible.

Proof via duality and complementary slackness (the hinted route)

Step 1: an LP that measures infeasibility. Introduce one extra variable t and consider

$$\text{(P)} \qquad \min_{x \in \mathbb{R}^n,\, t \in \mathbb{R}} \ t \qquad \text{s.t.} \qquad a_i^{\mathsf{T}} x - t \leq b_i, \quad i = 1, \ldots, m.$$

For fixed x the best t is $\max_i (a_i^{\mathsf{T}} x - b_i)$, the worst constraint violation at x. (P) is feasible (take any x and t large). Because $Ax \leq b$ has no solution, every feasible (x, t) has $t \geq \max_i (a_i^{\mathsf{T}} x - b_i) > 0$; so (P) is bounded below by 0. A feasible LP with bounded objective has an optimal solution $(x^\star, t^\star)$, and since $x^\star$ violates some inequality, $t^\star > 0$.

Step 2: the dual. Write the constraints as $-a_i^{\mathsf{T}} x + t \geq -b_i$ with multipliers $p_i \geq 0$. Both x and t are free, so the dual constraints are equalities:

$$\text{(D)} \qquad \max \ -b^{\mathsf{T}} p \qquad \text{s.t.} \qquad A^{\mathsf{T}} p = 0, \quad \mathbf{1}^{\mathsf{T}} p = 1, \quad p \geq 0.$$

By strong duality (D) has an optimal solution and its optimal value is $t^\star > 0$.

Step 3: a sparse optimal dual solution. (D) is an LP in *standard form*: nonnegative variables $p \in \mathbb{R}^m$ and $n+1 = 101$ equality constraints. Its feasible set contains no line (it lies in $p \geq 0$), so it has extreme points, and an LP with an optimal solution over such a polyhedron has an optimal solution $p^\star$ that is an extreme point, i.e. a basic feasible solution. In a basic feasible solution of a standard-form LP the columns of the equality matrix corresponding to positive variables are linearly independent; these columns live in $\mathbb{R}^{n+1}$, so

$$S = \{i : p_i^\star > 0\} \qquad \text{satisfies} \qquad |S| \leq n+1 = 101.$$

Step 4: the subsystem indexed by S is infeasible. Consider the LP (P_S) obtained from (P) by keeping only the constraints $i \in S$, and its dual (D_S), which is (D) with variables $(p_i)_{i \in S}$ only.

- $(x^\star, t^\star)$ is feasible for (P_S), since it satisfies *all* m constraints.
- $p_S^\star = (p_i^\star)_{i \in S}$ is feasible for (D_S): deleting zero entries does not change $\sum_i p_i^\star a_i = 0$ or $\sum_i p_i^\star = 1$.
- Complementary slackness holds for this pair. For the original optimal pair it gives $p_i^\star\,(b_i - a_i^\mathsf{T} x^\star + t^\star) = 0$ for all i, so every constraint $i \in S$ is tight at $(x^\star, t^\star)$; the conditions attached to the free variables are the dual equalities, which hold.

A primal-feasible and dual-feasible pair satisfying complementary slackness is optimal. Hence the optimal value of (P_S) is $t^\star > 0$: even the best x violates some inequality in S by $t^\star$. Therefore $\{x : a_i^\mathsf{T} x \leq b_i,\ i \in S\} = \emptyset$, with $|S| \leq 101$.

Shortcut. Weak duality alone finishes Step 4: $p_S^\star$ is feasible for (D_S) with value $-b_S^\mathsf{T} p_S^\star = t^\star$, so every feasible point of (P_S) has $t \geq t^\star > 0$. Complementary slackness adds the sharper information that the value is *exactly* $t^\star$ and that the constraints in S are all tight at $x^\star$.

Answer. There is always an infeasible subsystem of at most $n+1 = 101$ inequalities: the support of a basic optimal solution of the dual of the "minimise the worst violation" LP. Complementary slackness tells us precisely which inequalities matter: the ones that are tight at the point of least violation *and* carry positive dual weight. The other ≥ 399 inequalities are irrelevant to the infeasibility. Nothing is special about 500 and 100: for m inequalities in n variables the bound is $\min\{m, n+1\}$.

A second proof via Farkas' lemma (no optimisation needed)

Lemma (Farkas, inequality form)**.** $\{x : Ax \leq b\} = \emptyset$ *if and only if there is* $p \geq 0$ *with* $A^\mathsf{T} p = 0$ *and* $b^\mathsf{T} p < 0$.

(The "if" direction is immediate: a solution x would give $0 = p^\mathsf{T} Ax \leq p^\mathsf{T} b < 0$.) Given infeasibility, rescale the certificate so that $b^\mathsf{T} p = -1$. Then the polyhedron

$$Q = \{p \in \mathbb{R}^m :\ A^\mathsf{T} p = 0,\ b^\mathsf{T} p = -1,\ p \geq 0\}$$

is nonempty, in standard form, with $n+1$ equality constraints. A nonempty standard-form polyhedron has a basic feasible solution $\bar{p}$, with at most $n+1$ positive entries. Let S be its support. Then $\sum_{i \in S} \bar{p}_i a_i = 0$ and $\sum_{i \in S} \bar{p}_i b_i = -1 < 0$, so $\bar{p}_S$ is a Farkas certificate for the subsystem S, which is therefore infeasible by the easy direction of the lemma.

Why a BFS exists, constructively. If the columns $\binom{a_i}{b_i}$, $i \in \mathrm{supp}(p)$, are linearly dependent, pick $z \neq 0$ supported there with $\sum_i z_i \binom{a_i}{b_i} = 0$ and (flipping sign if needed) some $z_i > 0$. Moving to $p - \theta z$ keeps both equalities, and at $\theta = \min\{p_i/z_i : z_i > 0\}$ it keeps $p \geq 0$ while zeroing at least one more entry. Repeat until the columns are independent, which forces at most $n+1$ of them. This is the same argument that proves Carathéodory's theorem, and I ran it numerically below.

The bound $n+1$ is tight, and the geometric meaning

In $\mathbb{R}^n$ take the $n+1$ inequalities $x_j \geq 1$ $(j = 1, \ldots, n)$ and $\sum_j x_j \leq n - 1$. Together they are infeasible ($\sum_j x_j \geq n$), but dropping $x_k \geq 1$ leaves the feasible point $x = \mathbf{1} - e_k$, and dropping the sum constraint leaves $x = \mathbf{1}$. So no subset of size n is infeasible, and 101 cannot be replaced by 100. Figure 3 shows $n = 2$.

The contrapositive is **Helly's theorem for half-spaces**: *if every $n+1$ of a finite family of half-spaces in $\mathbb{R}^n$ have a common point, then all of them do.* Applications: an infeasible scheduling/resource model with thousands of constraints always has a "small" explanation of at most $n+1$ conflicting requirements (an *irreducible infeasible subsystem*, which solvers report for debugging); a linear classifier in $\mathbb{R}^n$ fails to separate a data set only if some $n+2$ points are already inseparable; minimax/Chebyshev fitting problems are determined by at most $n+1$ active data points.

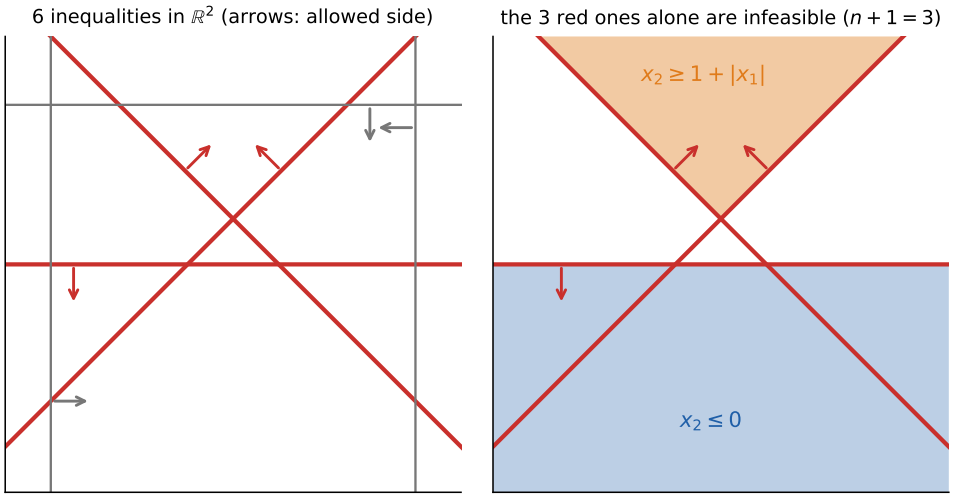


Figure 3: $n = 2$. Left: six inequalities with no common point. Right: three of them, $x_2 \leq 0$, $x_2 \geq x_1 + 1$, $x_2 \geq -x_1 + 1$, are already infeasible (the blue and orange regions are disjoint), with certificate $p = (1, \frac{1}{2}, \frac{1}{2})$: $\sum p_i a_i = 0$, $\sum p_i b_i = -1$. Any two of the three are feasible, so three is tight.

Numerical experiment

I generated a random system with $m = 500$, $n = 100$ (Gaussian A and b), which is infeasible with least worst-violation $t^\star = 1.1517$. Solving (D) with a simplex method returns a vertex with exactly 101 nonzero multipliers (Figure 4, left); the 101-row subsystem was confirmed infeasible by an independent LP, and removing *any* one of those 101 rows made it feasible, so this subset is minimal. Separately, starting from a dense Farkas certificate with 180 nonzeros, the reduction procedure above reached support 101 in 79 steps while preserving $A^\mathsf{T} p = 0$ and $b^\mathsf{T} p = -1$ to within 2×10^{-11} (Figure 4, right).

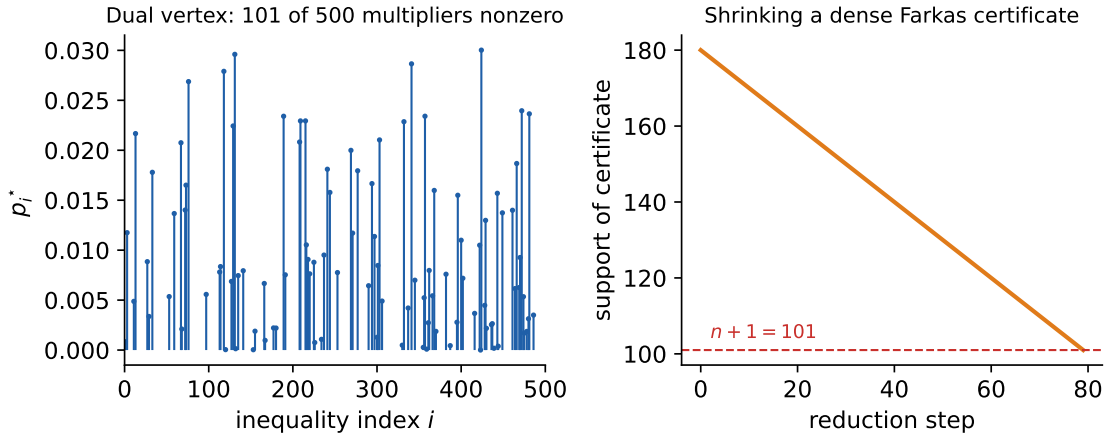


Figure 4: Left: the optimal dual vertex $p^\star$; its support identifies the infeasible subsystem. Right: support size during the constructive reduction of a dense certificate.

A caution from the experiment. My first feasibility test passed a zero objective to the solver and checked for the "infeasible" status flag; on the full 500×100 system the solver instead returned "status unknown", which my code misread as feasible. I replaced it with the phase-1 LP (P) itself, declaring infeasibility iff $t^\star > 0$; this LP is always feasible and (with a lower bound on t) bounded, so its answer is unambiguous. All results reported here use that test.

A Code and numerical checks

Figures were produced with `matplotlib` (`make_figures.py`); LPs were solved with SciPy's HiGHS interface.

```
"""Numerical checks for HW3: (1) primal/dual pair, (2) theorem of the alternative, (3) 101-subset
    result."""
import numpy as np
from scipy.optimize import linprog

rng = np.random.default_rng(0)
INF = None # linprog's notation for an infinite bound
free = lambda k: [(INF, INF)] * k

# ---------------------------------------------------------------------- Problem 1
print("=== Problem 1: primal and derived dual ===")
c = np.array([1.0, -1, 0, 0])
A = np.array([[2.0, 3, -1, 1], [3, 1, 4, -2], [-1, -1, 2, 1]])
b = np.array([0.0, 3, 6])
# primal: row1 <=, row2 >= (negated), row3 =; x1<=0, x2,x3>=0, x4 free
P = linprog(c, A_ub=np.vstack([A[0], -A[1]]), b_ub=[b[0], -b[1]], A_eq=A[2:3], b_eq=b[2:3],
            bounds=[(INF, 0), (0, INF), (0, INF), (INF, INF)], method="highs")
print("primal LP:", P.message.split(".")[0])
# dual: max 3p2+6p3 s.t. A1'p>=1, A2'p<=-1, A3'p<=0, A4'p=0, p1<=0, p2>=0, p3 free
D = linprog(-b, A_ub=np.vstack([-A[:, 0], A[:, 1], A[:, 2]]), b_ub=[-1, -1, 0],
            A_eq=A[:, 3][None, :], b_eq=[0], bounds=[(INF, 0), (0, INF), (INF, INF)], method="highs"
                )
print("dual LP :", D.message.split(".")[0])
x0, d = np.array([0.0, 0, 2, 2]), np.array([-1.0, 0, 1, -3])
```

```
for name, v, rhs in [("x0 ", x0, b), ("ray", d, 0 * b)]:
    r = A @ v
    ok = r[0] <= rhs[0] and r[1] >= rhs[1] and r[2] == rhs[2] and v[0] <= 0 and v[1] >= 0 and v[2]
        >= 0
    print(f"{name}= {v}: A{name.strip()} = {r}, c'{name.strip()} = {c @ v:+.0f}, satisfies
        constraints: {ok}")

# ------------------------------------------------------------------- Problem 2
print("\n=== Problem 2: (i) <=> (ii) on 2000 random integer matrices ===")
def stmt_i(M): # max x1 s.t. Mx>=0, x>=0, x1<=1 (the box keeps the LP bounded); (i) holds iff
    optimum is 0
    n = M.shape[1]; obj = np.zeros(n); obj[0] = -1
    r = linprog(obj, A_ub=-M, b_ub=np.zeros(M.shape[0]), bounds=[(0, 1)] + [(0, INF)] * (n - 1),
        method="highs")
    return abs(r.fun) < 1e-9
def stmt_ii(M): # find p>=0 with p'M<=0 and p'M1<=-1 (normalised form of the strict inequality)
    m, n = M.shape; rhs = np.zeros(n); rhs[0] = -1
    return linprog(np.zeros(m), A_ub=M.T, b_ub=rhs, bounds=[(0, INF)] * m, method="highs").status
        == 0
counts = {}
for _ in range(2000):
    M = rng.integers(-3, 4, size=(rng.integers(2, 7), rng.integers(2, 6))).astype(float)
    key = (stmt_i(M), stmt_ii(M)); counts[key] = counts.get(key, 0) + 1
print("both true:", counts.get((True, True), 0), "| both false:", counts.get((False, False), 0),
     "| disagreements:", counts.get((True, False), 0) + counts.get((False, True), 0))

# ------------------------------------------------------------------- Problem 3
print("\n=== Problem 3: 500 inequalities in 100 variables ===")
m, n = 500, 100
A3 = rng.standard_normal((m, n)); b3 = rng.standard_normal(m)
def infeasible(rows):
    """Phase-1 test: t* = min_x max_i (a_i'x - b_i), clipped below at -1. Infeasible <=> t* > 0.
    (A zero-objective feasibility LP is avoided: HiGHS can return 'status unknown' on it.)"""
    k = len(rows)
    r = linprog(np.r_[np.zeros(n), 1.0], A_ub=np.hstack([A3[rows], -np.ones((k, 1))]), b_ub=b3[rows
        ],
                bounds=free(n) + [(-1, INF)], method="highs")
    assert r.status == 0, r.message
    return r.fun > 1e-7
print("full system infeasible:", infeasible(np.arange(m)))

# (a) duality route. Dual of min t s.t. a_i'x - t <= b_i is max -b'p s.t. A'p=0, 1'p=1, p>=0 (101
    equalities)
Aeq = np.vstack([A3.T, np.ones((1, m))]); beq = np.r_[np.zeros(n), 1.0]
dual = linprog(b3, A_eq=Aeq, b_eq=beq, bounds=[(0, INF)] * m, method="highs-ds") # simplex => a
    vertex
p = dual.x; S = np.flatnonzero(p > 1e-9)
print(f"(a) dual optimum t* = {-dual.fun:.4f} > 0; vertex support |S| = {len(S)} (bound n+1 = {n +
    1})")
print(f" certificate: ||sum p_i a_i|| = {np.linalg.norm(A3[S].T @ p[S]):.1e}, sum p_i b_i = {p[S] @
     b3[S]:.4f} < 0")
print(" subsystem S infeasible (independent LP check):", infeasible(S))
drop = sum(infeasible(np.delete(S, k)) for k in range(len(S)))
print(f" dropping any single row of S: still infeasible in {drop} of {len(S)} cases (so S is
    minimal)")

# (b) constructive route: start from a DENSE Farkas certificate and shrink its support (
    Caratheodory-style)
verts = [linprog(b3 + 0.3 * rng.standard_normal(m), A_eq=Aeq, b_eq=beq, bounds=[(0, INF)] * m,
```

```
    method="highs-ds").x
        for _ in range(6)]
q = np.mean(verts, axis=0); q /= -(b3 @ q) # normalise so that b'q = -1
print(f"(b) dense certificate: support {np.sum(q > 1e-12)}, ||A'q|| = {np.linalg.norm(A3.T @ q):.1e
    }, b'q = {b3 @ q:.4f}")
history = [int(np.sum(q > 1e-12))]
while True:
    T = np.flatnonzero(q > 1e-12)
    M = np.vstack([A3[T].T, b3[T][None, :]]) # 101 x |T|
    if np.linalg.matrix_rank(M) == len(T): break # columns independent: cannot shrink further
    z = np.linalg.svd(M)[2][-1] # a null vector: M z = 0
    if z.max() <= 0: z = -z
    pos = z > 1e-14
    theta = np.min(q[T][pos] / z[pos]) # largest step keeping q >= 0
    q[T] = np.maximum(q[T] - theta * z, 0); q[q < 1e-12] = 0
    history.append(int(np.sum(q > 0)))
T = np.flatnonzero(q > 0)
print(f" after {len(history) - 1} reduction steps: support {len(T)}, ||A'q|| = {np.linalg.norm(A3.T
    @ q):.1e}, b'q = {b3 @ q:.4f}")
print(" reduced subsystem infeasible (independent LP check):", infeasible(T))
np.savez("p3_data.npz", p=p, history=np.array(history))
```

Output:

```
=== Problem 1: primal and derived dual ===
primal LP: The problem is unbounded
dual LP : The problem is infeasible
x0 = [0. 0. 2. 2.]: Ax0 = [0. 4. 6.], c'x0 = +0, satisfies constraints: True
ray= [-1. 0. 1. -3.]: Aray = [-6. 7. 0.], c'ray = -1, satisfies constraints: True

=== Problem 2: (i) <=> (ii) on 2000 random integer matrices ===
both true: 1140 | both false: 860 | disagreements: 0

=== Problem 3: 500 inequalities in 100 variables ===
full system infeasible: True
(a) dual optimum t* = 1.1517 > 0; vertex support |S| = 101 (bound n+1 = 101)
    certificate: ||sum p_i a_i|| = 2.6e-11, sum p_i b_i = -1.1517 < 0
    subsystem S infeasible (independent LP check): True
    dropping any single row of S: still infeasible in 0 of 101 cases (so S is minimal)
(b) dense certificate: support 180, ||A'q|| = 1.8e-11, b'q = -1.0000
    after 79 reduction steps: support 101, ||A'q|| = 1.8e-11, b'q = -1.0000
    reduced subsystem infeasible (independent LP check): True
```